

### Quick Activity 1: School Supplies Calculator

A notebook has a prize of ₱45, and a pen has a prize of ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the **total prize** and **change**.

#### Decomposition

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase

| INPUT                                           | PROCESS                                                                                    | OUTPUT                 |
|-------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------|
| Number of notebooks,<br>number of pens, payment | Calculate notebook prize,<br>calculate pen prize, add for<br>total prize, calculate change | Total prize and change |

#### Pseudocode

START

```
READ notebook, pen, payment
notebookPrize ← notebook × 45
penPrize ← pen × 12
totalPrize ← notebookPrize + penPrize
change ← payment - totalPrize
DISPLAY totalPrize
DISPLAY change
```

END

## Dry Run Calculations

### Test Case 1

#### Input:

- Notebook: 3
- Pen = 2
- Payments = ₱200

| Step | Variable       | Calculation   | Solution |
|------|----------------|---------------|----------|
| 1    | Notebook Prize | $3 \times 45$ | ₱135     |
| 2    | Pen Prize      | $2 \times 12$ | ₱24      |
| 3    | Total Prize    | $135 + 24$    | ₱159     |
| 4    | Change         | $200 - 159$   | ₱41      |

#### Output

Total Cost = ₱159

Change = ₱41

### Test Case 2

#### Input:

- Notebook = 2
- Pen = 3
- Payment = ₱150

| Step | Variable       | Calculation   | Solution |
|------|----------------|---------------|----------|
| 1    | Notebook Prize | $2 \times 45$ | ₱90      |
| 2    | Pen Prize      | $3 \times 12$ | ₱36      |
| 3    | Total Prize    | $90 + 36$     | ₱126     |
| 4    | Change         | $150 - 126$   | ₱24      |

#### Output

Total Cost = ₱126

Change = ₱24